

- 8** The scientific document you have studied is adapted from an article in *National Geographic*: How personalized medicine is transforming your health care.

Use the information from the scientific document and your own knowledge to answer the following questions.

- (a) Explain what is meant by the phrase 'gene variants' (paragraph 4).

(2)

- (b) Explain why 'tumors ... riddled with different mutations' are good candidates for immunotherapies (paragraph 6).

(4)

(c) Explain why a CT scan was used to show that Judy Perkins was free of tumours (paragraph 10 and Figure 1).

(2)

(d) Checkpoints are a type of protein that prevent the activation of immune cells.

Explain why a checkpoint inhibitor was given with the lymphocytes in the immunotherapy used to treat Judy Perkins (paragraph 10).

(2)

(e) Explain how 'a gene variant that produces a defective form of an enzyme' reduces the effectiveness of clopidogrel (paragraph 16).

(4)

(f) Researchers have used stem cells to produce spinal cord tissue from individuals with amyotrophic lateral sclerosis.

These functioning tissues contain motor neurones and blood vessels.

Suggest why blood vessels are required to form functioning spinal cord tissue (paragraph 31).

(3)

[illegible]

(g) Suggest how 'a brew of growth factors and other proteins' can stimulate induced pluripotent stem cells to produce a functioning tissue (paragraph 33).

(3)

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(Total for Question 8 = 20 marks)

TOTAL FOR PAPER = 90 MARKS